# Rohin Rajagopal

ms23btech11023@iith.ac.in | rohin.rajgopal@gmail.com | +91 9972324354 | LinkedIn

## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 7.71 |
| Narayana E-Techno School | Class XII (CBSE) | 2023 | 95.40% |
| Silver Oaks International School | Class X (CBSE) | 2021 | 90.20% |

## ACHIEVEMENTS

- Represented **India as 1 of 32** selected members at **YMUN XLV (2019)**, held at **Yale University, New Haven, Connecticut**, alongside **2,000+ delegates from around the world.**

## PROJECTS

**Text Summarizer Encoder Decoder Bahdanau Attention:**

- Developed an LSTM-based Encoder-Decoder model with Bahdanau (additive) attention and custom-trained word embeddings, using TimeDistributed layer to enhance token-level context in abstractive summarization
- Leveraged the Inshorts News Dataset to train and validate the model, yielding context-sensitive news summaries

Custom Memory Allocator:

- **Implemented an mmap-backed replacement for malloc, free, calloc, and realloc in C++17**, using **boundary tags, 16 segregated free lists, block splitting, and guaranteed 16-byte alignment**.
- **Engineered O(1) bidirectional block coalescing** using boundary tags and doubly linked free lists, while maintaining heap invariants such as no adjacent free blocks and one-list-per-free-block membership.
- **Built a custom heap-verification framework and 23 GoogleTest cases** covering header/footer consistency, alignment, free-list integrity, coalescing, overflow detection, and randomized allocator workloads.

Fantasy Football API and backend:

- **Built a 9-endpoint REST API** using **Python 3.12, FastAPI, SQLAlchemy 2.0, and PostgreSQL 16**, implementing JWT authentication, Argon2id password hashing, a **7-table relational schema**, and Alembic migrations; containerized services with Docker Compose and automated migration, linting, and testing through **GitHub Actions**.
- **Engineered concurrency-safe player transfers** using a **READ COMMITTED transaction with SELECT ... FOR UPDATE row locking**, acquiring locks in a fixed ID order and re-reading budget/squad state under lock; implemented **idempotent retries** with an Idempotency-Key protocol and UNIQUE(manager_id, key) constraint to prevent duplicate concurrent transfers.
- **Built 77 Pytest cases** against a real PostgreSQL instance, including a **10-thread concurrent test harness** using independent database connections to verify exact budget reconciliation; reproduced the **lost-update anomaly** when row locking was removed, validating the concurrency tests.

FIFA WC match predictor and tournament simulator:

- Implemented a **Dixon-Coles football prediction model** in Python (NumPy/SciPy), fitting **504 parameters across 251 teams** via weighted maximum likelihood with a **hand-derived analytical gradient and L-BFGS-B optimization**.
- Achieved **68.8% accuracy and 0.819 log-loss** on **96 held-out matches** vs. **1.099 uniform baseline**; simulated **10,000 tournaments in <20s** using cached probability grids and validated model, probability and simulation invariants with a **15-case Pytest suite**.

## POSITIONS OF RESPONSIBILITY

**Operations Head, E-Cell IIT Hyderabad | AY 2025-2026**

- **Led logistics and vendor operations** for E-Summit, coordinating **50+ team members** and managing execution for **2,000+ attendees.**
- Coordinated with **institute administration** to secure **approvals for venues, budgets, and event logistics,** ensuring timely execution across multiple initiatives.

- Acquired **1.2L in sponsorship** from food vendors through **targeted outreach and partnership pitches.**

**Public Relations Coordinator, International Relations Cell | AY 2024-2025**

- Engaged **100+ Japanese universities and companies** to establish institutional collaborations for Japan Week under the **JICA Program.**
- Developed and executed **campus-wide outreach campaigns** for Japan Week, promoting **international career and job opportunities** in Japan.

## SKILLS

Skills: Languages: Python, C++, C, SQL

Backend: FastAPI, REST APIs, SQLAlchemy, JWT auth

Databases: PostgreSQL, MySQL

Tools: Git, GitHub, Docker, Linux, pytest, Postman

ML & Data: Scikit-learn, TensorFlow, NLTK, NumPy, Pandas, Matplotlib

Domain: Options trading (Greeks, IV, hedging), Power BI, Excel

Soft Skills: Communication, conflict resolution, active listening, negotiation and persuasive pitching

## EXTRACURRICULAR

- Completed **Grade 4 Classical Drums** certification through **Trinity College London**.
- Member of the National Service Scheme.
- Raised funds to support Animal Welfare Society initiatives.